

LAB 8 Questions

1. You are provided with a synthetic dataset of 20 samples representing a binary classification problem. The feature x is a continuous variable, and the label $y \in \{0, 1\}$ represents the class.

- **Dataset:**

- **X:** [0.5, 0.8, 1.2, 1.5, 2.0, 2.4, 2.8, 3.1, 3.5, 4.2, 5.2, 5.5, 5.8, 6.2, 6.5, 7.1, 7.8, 8.2, 8.9, 9.5]
- **Y:** [0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]

- Assuming following two models for prediction system

- **Linear Model :**

$$h_{(m,c)}^1(x) = mx + c$$

- **Logistic Model :**

$$h_{(m,c)}^2(x) = \frac{1}{1 + e^{-(mx+c)}}$$

- Compare the behavior of the **Mean Squared Error (MSE)** loss function when applied to two different hypothesis models. For both models, assume the bias (intercept) c is fixed at 0. **The Cost Function:**

$$J_{\theta} = \frac{1}{n} \sum_{i=1}^n (h(x^{(i)}) - y^{(i)})^2$$

- Write a script to calculate J_{θ} for both models over a range of slope values $m \in [-10, 10]$. Generate a single figure with two subplots.
 - **Subplot A:** J_{θ} vs. m for the Linear Model.
 - **Subplot B:** J_{θ} vs. m for the Logistic Model.
- Provide a brief written response addressing the following:
 - Which plot represents a **convex** function?
 - In the Logistic-MSE plot, identify any regions where the gradient $\frac{dJ}{dm}$ is near zero (plateaus). Explain how these regions would impact optimizers.

2. Assume the Error function to be **Negative Log-Likelihood (NLL)**,

- defined as:

$$J_{\theta} = -\frac{1}{n} \sum_{i=1}^n \left[y^{(i)} \ln(h(x^{(i)})) + (1 - y^{(i)}) \ln(1 - h(x^{(i)})) \right]$$

- For the logistic model $h_{(m,c)}^2(x)$ given as above, plot the graph for error function **NLL** for $m \in \{-10, 10\}$ keeping $c = 0$.
- What advantage it will give to the optimizer algorithm?

3. Use optimizers to solve for the model parameters.

- Use simulated annealing to solve for m and c for the model $h_{(m,c)}^2(x)$ with MSE as error function (to be minimized). Use stopping condition as when the improvement is less than 10^{-5} stop.
- Use hill climbing to solve for m and c for the model $h_{(m,c)}^2(x)$ with **NLL** as error function (to be minimized.)
- Are you able to reach to same value of m and c in both cases ?
- Provide reasoning behind the choice of optimizer type.